

AI-POWERED EMAIL SPAM DETECTION SYSTEM (CYBERINBOX)

Summer Training Project Submitted in Partial Fulfillment for the Award of
the Degree of Bachelor of Technology in Computer Science and
Engineering



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IBM SKILLSBUILD AI INTERNSHIP

JULY 2026

CERTIFICATE

This is to certify that the Summer Training Project entitled "**AI-Powered Email Spam Detection System (CyberInbox)**" submitted by **AVJEET KUMAR** in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering under Dr. A. P. J. Abdul Kalam Technical University, Lucknow, has been carried out successfully at **IBM SkillsBuild Virtual Internship Program**

The work embodied in this report is original and has been completed under my supervision during the IBM SkillsBuild AI Internship.

Signature of the Guide:

Mr. Shubham Pathak

CANDIDATE’S DECLARATION

I hereby declare that the project entitled **”AI-Powered Email Spam Detection System (CyberInbox)”** submitted in partial fulfillment of the requirements for the award of Bachelor of Technology in Computer Science and Engineering is an original work carried out by me during the IBM SkillsBuild AI Internship under the guidance of **Mr. Shubham Pathak, Technical Mentor, IBM SkillsBuild Virtual Internship Program.**

I further declare that this work has not been submitted either wholly or partially for the award of any degree, diploma or any other academic qualification.

Signature of Student

AVJEET KUMAR

ACKNOWLEDGEMENT

I express my sincere gratitude to the Department of Computer Science and Engineering, **IBM SkillsBuild** , for providing the opportunity to undertake this Summer Training Project under the IBM SkillsBuild AI Internship.

I would like to express my heartfelt thanks to my guide, **Mr.Shubham Pathak**, for her continuous encouragement, valuable suggestions, and expert guidance throughout the project. Her support motivated me to complete this work successfully.

I also thank the IBM SkillsBuild AI Internship program for providing an excellent learning platform to explore Artificial Intelligence and Machine Learning concepts through practical implementation.

Finally, I extend my gratitude to all faculty members, friends, and family members whose constant support and encouragement made the successful completion of this project possible.

ABSTRACT

Electronic mail remains one of the most widely used communication platforms across businesses, educational institutions, and individuals. However, the rapid increase in unsolicited and malicious emails has created significant cybersecurity challenges. Spam emails are frequently used to distribute phishing links, malware, ransomware, and fraudulent advertisements, making automatic spam detection an essential requirement.

This project presents an Artificial Intelligence based Email Spam Detection System named **CyberInbox**. The system utilizes Natural Language Processing (NLP) and Machine Learning techniques to classify incoming emails as Spam or Ham (Legitimate). Text preprocessing techniques including lowercase conversion, punctuation removal, stop-word elimination, URL removal, email removal, and tokenization are applied before feature extraction using TF-IDF Vectorization.

The processed features are used to train a Logistic Regression classifier on the SpamAssassin public dataset. The trained model is deployed using the FastAPI framework, while a modern user interface is developed using React.js, Vite, and Tailwind CSS. The application also provides prediction confidence, dataset statistics, and model performance metrics through an interactive dashboard.

Experimental evaluation demonstrates high classification performance with excellent Accuracy, Precision, Recall, and F1-Score, making the proposed solution suitable for practical email filtering applications.

The project highlights the integration of Artificial Intelligence with modern web technologies to develop an efficient, scalable, and user-friendly cybersecurity solution capable of reducing spam-related threats in real-world environments.

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Chapter 1

Introduction

1.1 *Introduction to the Project*

Email has become one of the most widely used communication mediums for individuals, businesses, educational institutions, and government organizations. Millions of emails are exchanged every day for personal and professional communication. Along with legitimate emails, the number of unsolicited and malicious emails, commonly known as spam, has increased significantly over the years. Spam emails not only consume storage and network resources but also serve as a medium for phishing attacks, malware distribution, ransomware delivery, financial fraud, and identity theft.

Traditional rule-based spam filters depend on manually created rules and predefined keywords. Although these methods are useful, they often fail to detect newly emerging spam patterns and sophisticated phishing techniques. With the advancement of Artificial Intelligence (AI) and Machine Learning (ML), intelligent spam detection systems have become capable of learning hidden patterns directly from email datasets and providing more accurate predictions.

This project presents an AI-powered Email Spam Detection System named **CyberInbox**. The system automatically classifies emails into two categories: **Spam** and **Ham (Legitimate Email)** using Natural Language Processing (NLP) and Machine Learning techniques. The application has been developed using a Logistic Regression classifier trained on the SpamAssassin public dataset. The backend is implemented using FastAPI while the frontend is developed using React.js, Vite, and Tailwind CSS to provide a modern and interactive user interface.

1.2 *Problem Statement*

Electronic mail is one of the primary targets of cybercriminals due to its widespread usage. Every day, organizations receive thousands of spam emails containing phishing links, fake advertisements, malicious attachments, cryptocurrency scams, and ransomware payloads. These emails cause financial losses, compromise sensitive information, and reduce user productivity.

Existing spam filtering mechanisms often depend on manually maintained rules, making them ineffective against continuously evolving spam techniques. There is a need for an intelligent system capable of automatically identifying spam emails based on learned textual patterns rather than static rules.

The objective of this project is to develop an Artificial Intelligence-based spam email detection system capable of accurately distinguishing spam emails from legitimate emails using Machine Learning algorithms.

1.3 *Objectives*

The primary objectives of this project are:

- To study different types of spam emails and their characteristics.
- To preprocess raw email data using Natural Language Processing techniques.
- To extract meaningful textual features using TF-IDF Vectorization.
- To train a Machine Learning model for spam email classification.
- To evaluate model performance using Accuracy, Precision, Recall, and F1-Score.
- To develop a REST API using FastAPI for real-time predictions.
- To build a responsive web application using React.js and Tailwind CSS.
- To display prediction confidence and model statistics through an interactive dashboard.
- To provide a scalable and user-friendly AI-powered spam detection platform.

1.4 *Scope of the Project*

The proposed system focuses on detecting spam emails using Artificial Intelligence techniques. The application accepts email text from users through a web interface and predicts whether the email belongs to the Spam or Ham category.

The scope of the project includes:

- Automatic spam email classification.
- Real-time prediction through REST API.
- Natural Language Processing for text cleaning.
- Machine Learning model deployment.
- Interactive visualization of prediction confidence.
- Dashboard showing dataset statistics and model performance.
- Easy deployment for educational and research purposes.

Although the current implementation uses Logistic Regression, the architecture allows future integration of advanced Deep Learning models such as LSTM, BERT, RoBERTa, and Transformer-based classifiers.

1.5 *Technologies Used*

The CyberInbox application has been developed using the following technologies:

Technology	Purpose
Python	Machine Learning Development
FastAPI	Backend REST API
React.js	Frontend Development
Vite	React Build Tool
Tailwind CSS	User Interface Design
Scikit-learn	Machine Learning Library
Pandas	Dataset Processing
NumPy	Numerical Computing
NLTK	Natural Language Processing
TF-IDF Vectorizer	Feature Extraction
Logistic Regression	Email Classification
Joblib	Model Serialization
Axios	API Communication

1.6 *Project Workflow*

The complete workflow of CyberInbox consists of the following stages:

1. Collection of SpamAssassin Email Dataset.
2. Data Cleaning and Text Preprocessing.
3. Tokenization and Stop-word Removal.
4. TF-IDF Feature Extraction.
5. Logistic Regression Model Training.
6. Model Evaluation.
7. Saving Trained Model.
8. FastAPI Backend Development.
9. React Frontend Development.
10. Real-time Email Prediction.

The workflow enables end-to-end spam detection from raw email text to final prediction with confidence score.

Chapter 2

Literature Review

2.1 *Introduction*

Email spam detection has been an active area of research for more than two decades due to the rapid increase in unsolicited and malicious emails. Researchers have proposed various approaches ranging from rule-based filtering systems to modern Artificial Intelligence and Deep Learning techniques. Earlier spam filters relied on manually created rules and blacklists, but these approaches required continuous updates and often failed against newly emerging spam patterns.

With the advancement of Machine Learning, automatic spam classification has become significantly more accurate. Algorithms such as Naive Bayes, Support Vector Machine (SVM), Decision Tree, Random Forest, Logistic Regression, and Neural Networks have demonstrated excellent performance in classifying spam emails by learning textual patterns directly from datasets.

Natural Language Processing (NLP) techniques have further improved spam detection by transforming raw textual data into meaningful numerical representations. Methods such as Tokenization, Stop-word Removal, Stemming, Lemmatization, and TF-IDF Vectorization are widely used before model training.

Recent studies also explore Deep Learning architectures such as LSTM, GRU, BERT, and Transformer models, which provide higher accuracy but require larger datasets and computational resources.

Considering the balance between performance, simplicity, and computational efficiency, Logistic Regression combined with TF-IDF feature extraction has been selected for this project.

2.2 *Existing Systems*

Traditional spam filtering systems generally use one or more of the following approaches:

- Keyword-based Filtering
- Blacklist and Whitelist Filtering
- Rule-based Spam Detection
- Signature-based Detection
- Bayesian Spam Filtering

Although these methods successfully identify common spam emails, they suffer from several limitations.

- Require frequent manual updates.
- Unable to detect newly generated spam.
- High false positive rates.
- Difficult to scale for large organizations.
- Cannot learn automatically from new email patterns.

These limitations motivate the use of Machine Learning techniques for intelligent spam detection.

2.3 *Proposed System*

The proposed system, **CyberInbox**, utilizes Artificial Intelligence and Natural Language Processing to automatically classify emails into Spam and Ham categories.

The complete system consists of four major modules:

- Email Data Preprocessing
- Feature Extraction using TF-IDF
- Logistic Regression Classification
- FastAPI based Prediction Service with React Frontend

The application provides users with an easy-to-use interface where email content can be pasted or uploaded for instant analysis.

Chapter 3

Dataset Description

3.1 *SpamAssassin Dataset*

The SpamAssassin Public Corpus is one of the most widely used benchmark datasets for spam email classification research. It contains a collection of legitimate (Ham) and Spam emails collected from real-world email traffic.

The dataset includes various categories of emails such as promotional advertisements, phishing attempts, newsletters, personal communications, and business emails.

Each email is labeled into one of the following classes:

Label	Description
0	Ham (Legitimate Email)
1	Spam Email

The dataset was cleaned and converted into CSV format before training the Machine Learning model.

3.2 *Dataset Preprocessing*

Raw email data cannot be directly used for Machine Learning because it contains unnecessary symbols, HTML tags, URLs, email addresses, punctuation, numbers, and other noisy information.

The following preprocessing steps were applied:

1. Conversion to Lowercase
2. Removal of HTML Tags
3. Removal of URLs
4. Removal of Email Addresses
5. Removal of Numbers
6. Removal of Special Characters
7. Removal of Stop Words
8. Tokenization
9. Whitespace Normalization

These preprocessing steps improve feature quality and reduce noise within the dataset.

Chapter 4

Methodology

4.1 *Natural Language Processing*

Natural Language Processing (NLP) enables computers to understand and analyze human language. Since emails consist of textual information, NLP techniques are essential before Machine Learning algorithms can process them.

The cleaned text is transformed into structured numerical data suitable for classification.

4.2 *TF-IDF Feature Extraction*

Term Frequency-Inverse Document Frequency (TF-IDF) converts textual information into numerical vectors.

TF measures how frequently a word appears within an email, whereas IDF reduces the importance of very common words appearing in many documents.

The TF-IDF representation provides meaningful numerical features while minimizing the influence of common but less informative words.

This representation significantly improves classification performance compared to simple word counting techniques.

4.3 *Machine Learning Model*

The Logistic Regression algorithm has been selected as the classification model due to its simplicity, efficiency, and excellent performance on binary classification problems.

The model learns a decision boundary separating Spam and Ham emails based on TF-IDF feature vectors.

Reasons for selecting Logistic Regression include:

- High classification accuracy
- Fast training
- Low computational complexity
- Suitable for binary classification
- Easy deployment
- Excellent interpretability

The trained model is stored using the Joblib library and later loaded by the FastAPI backend for real-time prediction.

4.4 *Model Training*

The complete training pipeline consists of the following steps:

1. Load SpamAssassin Dataset
2. Clean Email Text
3. Generate TF-IDF Features
4. Split Dataset into Training and Testing Sets
5. Train Logistic Regression Model
6. Evaluate Performance
7. Save Model using Joblib

The trained classifier is capable of predicting whether an unseen email belongs to the Spam or Ham category.

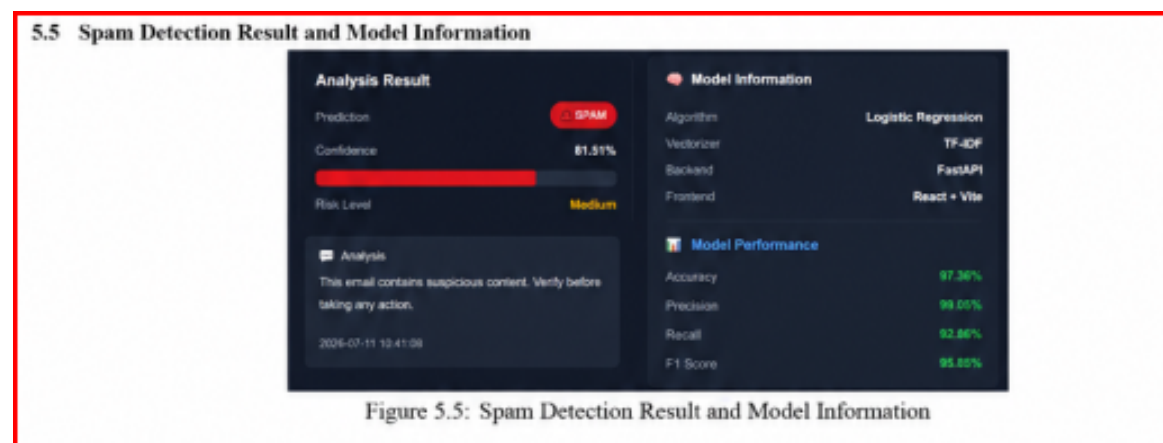


Figure 4.1: Prediction Summary and Model Information

4.5 Performance Evaluation

The performance of the proposed model is evaluated using standard Machine Learning metrics.

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

These metrics provide a comprehensive evaluation of the classifier's prediction capability and ensure that the developed model performs effectively on unseen email data.

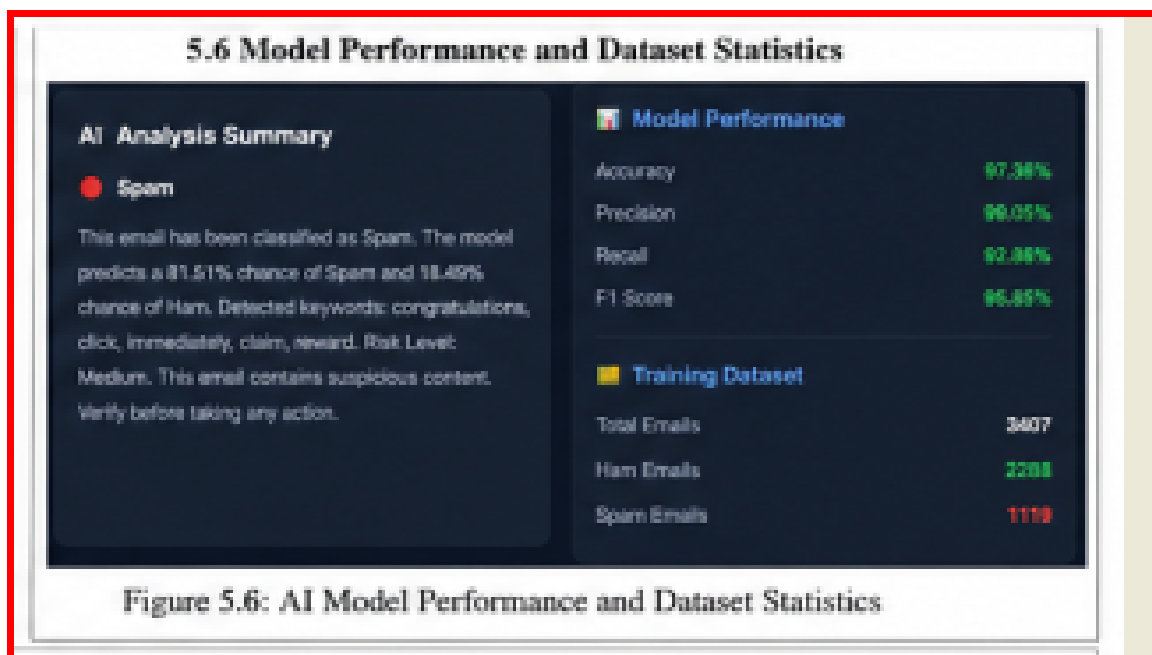


Figure 4.2: Model Performance Metrics

Chapter 5

System Design and Implementation

5.1 *System Architecture*

The CyberInbox system follows a client-server architecture consisting of four major components:

1. User Interface (React.js)
2. FastAPI Backend
3. Machine Learning Model
4. SpamAssassin Dataset

The workflow begins when the user enters or uploads an email through the web interface. The email is sent to the FastAPI backend, where preprocessing and feature extraction are performed. The processed text is transformed using the trained TF-IDF Vectorizer and passed to the Logistic Regression classifier. The prediction result and confidence score are returned to the frontend and displayed to the user.

Workflow

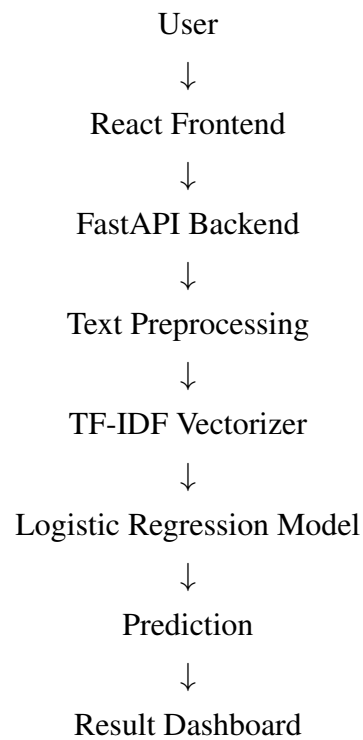


Figure 5.8: CyberInbox Project Directory Structure

Figure 5.1: Overall Architecture of the CyberInbox System

5.2 *Backend Development*

The backend of CyberInbox has been developed using FastAPI.

FastAPI provides a lightweight, high-performance REST API framework suitable for Machine Learning deployment.

The backend performs the following operations:

- Receives email text from frontend
- Cleans and preprocesses the text
- Loads trained TF-IDF Vectorizer
- Loads Logistic Regression model
- Generates prediction
- Calculates confidence score
- Returns prediction in JSON format

The REST API makes the Machine Learning model easily accessible to any frontend application.

5.3 *Frontend Development*

The frontend has been developed using React.js together with Vite and Tailwind CSS.

The user interface consists of multiple reusable components that improve usability and maintainability.

Major components include:

- Navigation Bar
- Hero Section
- Email Analyzer
- Result Card
- Confidence Gauge
- Statistics Dashboard
- Dataset Charts
- Spam Keyword Visualization

The responsive interface enables users to analyze emails efficiently on different screen sizes.

5.1 CyberInbox Home Page

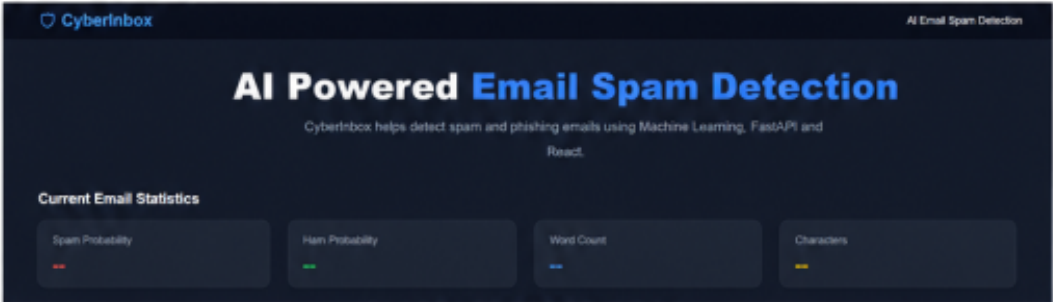


Figure 5.1: CyberInbox Home Page

Figure 5.2: CyberInbox Home Page

5.2 Email Analysis Interface

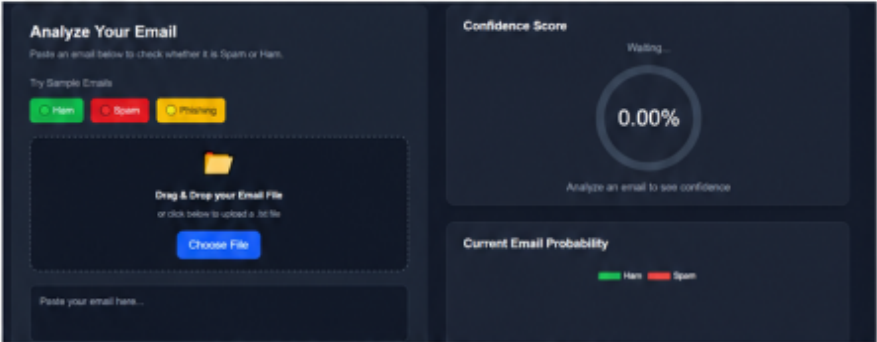


Figure 5.2: Email Upload and Analysis Interface

Figure 5.3: Email Analysis Interface

5.4 *Prediction Process*

The prediction process consists of the following steps:

1. User enters email text.
2. Email is sent to FastAPI API.
3. Text preprocessing is performed.
4. TF-IDF feature vector is generated.
5. Logistic Regression predicts Spam or Ham.
6. Prediction confidence is calculated.
7. Result is displayed on dashboard.

This process takes only a fraction of a second, making the application suitable for real-time spam detection.

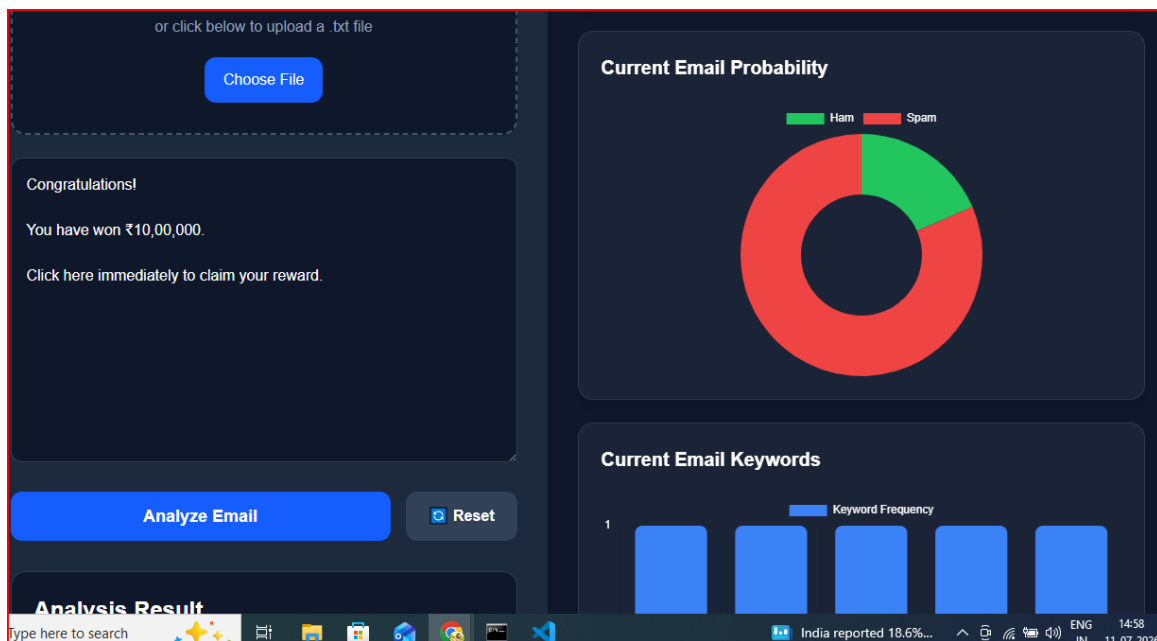


Figure 5.4: Spam Detection Result

Chapter 6

Experimental Results

6.1 *Model Performance*

The trained Logistic Regression classifier achieved excellent classification performance on the testing dataset.

Metric	Value
Accuracy	97.36%
Precision	99.05%
Recall	92.86%
F1-Score	95.85%

These results indicate that the developed classifier successfully distinguishes spam emails from legitimate emails with high accuracy.

6.2 Output Screens

The following screenshots demonstrate the working of CyberInbox.

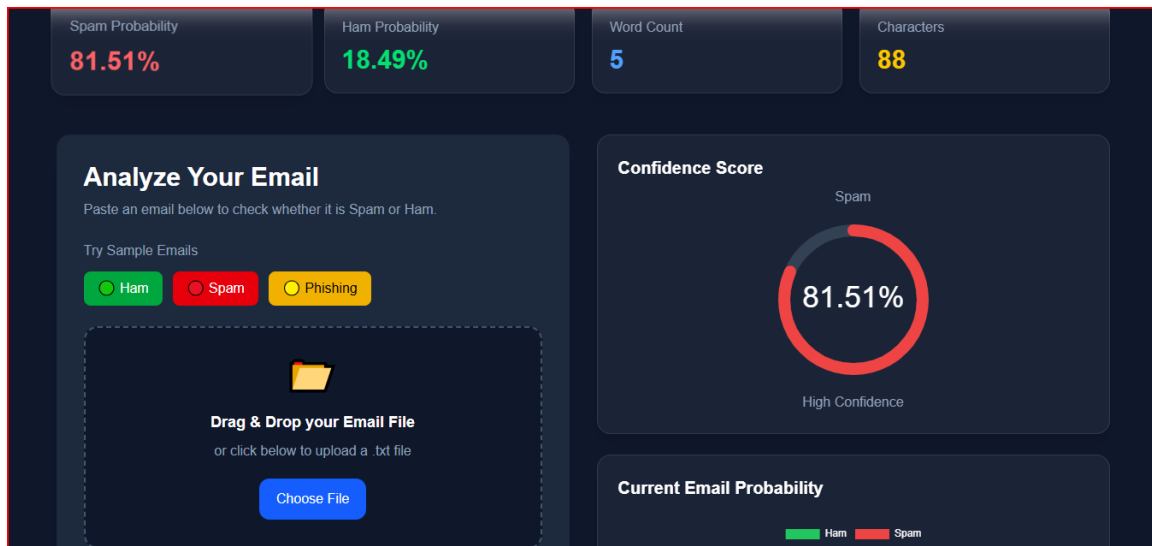


Figure 6.1: Home Page

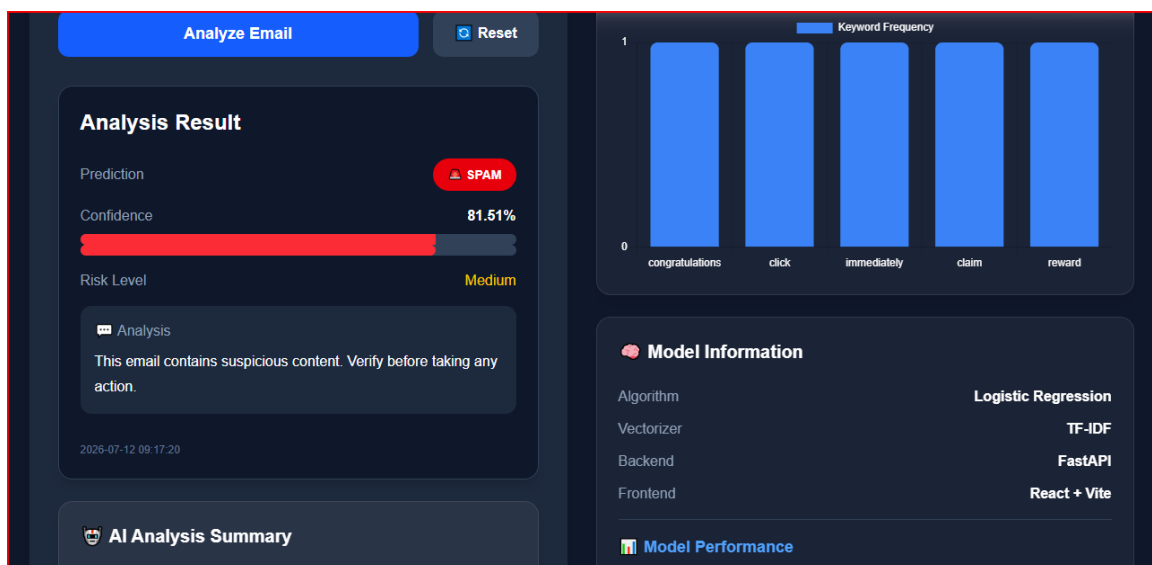


Figure 6.2: Spam Prediction

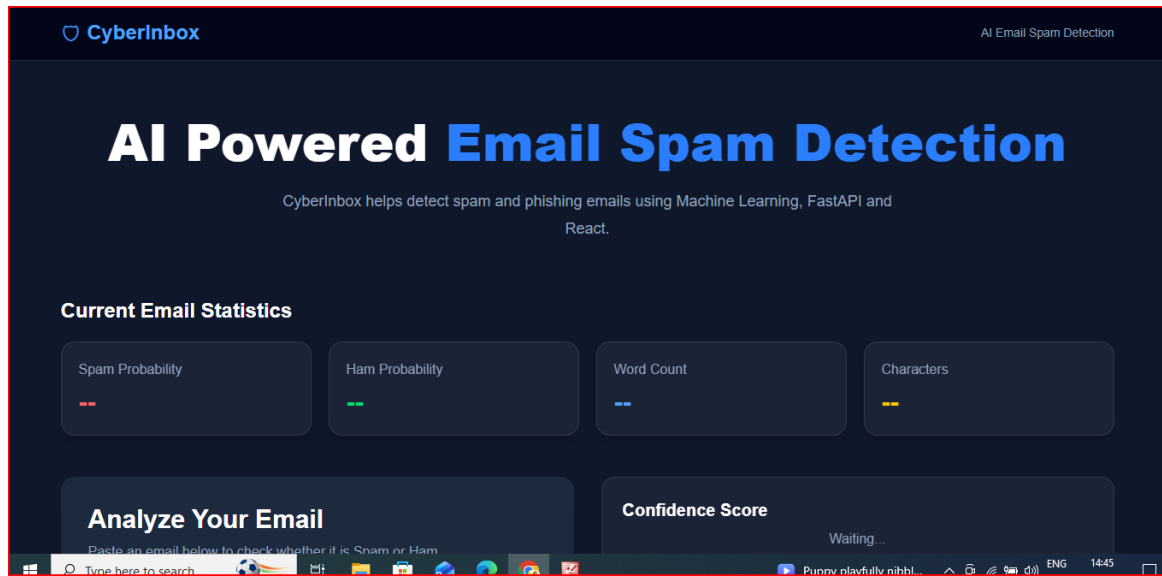


Figure 6.3: Dashboard

6.3 *Advantages*

- High prediction accuracy
- Fast response time
- User-friendly interface
- Real-time spam detection
- REST API integration
- Scalable architecture
- Lightweight Machine Learning model
- Easy deployment

6.4 *Limitations*

Despite its strong performance, the current implementation has some limitations.

- Works only on English emails.
- Image-based spam is not detected.
- Attachments are not analyzed.
- Requires retraining for new spam patterns.
- Uses only textual information.

Chapter 7

Future Scope

The proposed system can be enhanced in several directions.

- Integration with Gmail and Outlook.
- Deep Learning models such as LSTM and BERT.
- Multi-language spam detection.
- Image and attachment analysis.
- Phishing URL detection.
- Browser extension.
- Mobile application.
- Cloud deployment.
- Real-time enterprise email gateway.

Chapter 8

Conclusion

CyberInbox demonstrates the successful application of Artificial Intelligence and Natural Language Processing for spam email classification.

The project combines TF-IDF feature extraction with the Logistic Regression algorithm to build an efficient spam detection model capable of accurately distinguishing spam from legitimate emails.

The developed web application integrates Machine Learning with FastAPI and React.js to provide a complete end-to-end solution for real-time email analysis.

The obtained performance metrics demonstrate that Logistic Regression remains an effective algorithm for text classification while maintaining low computational complexity.

Overall, the project fulfills its objective of developing a practical, scalable, and user-friendly AI-powered spam email detection system suitable for educational and real-world applications.

Chapter 9

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